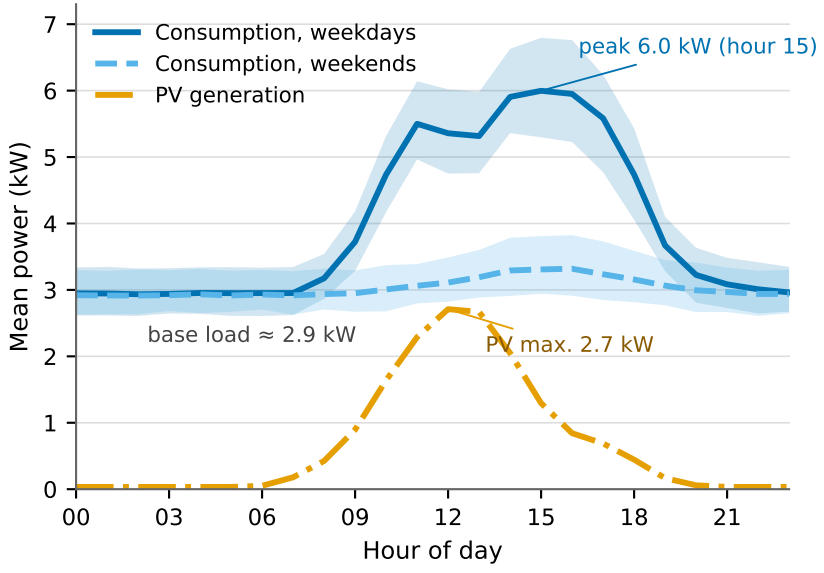
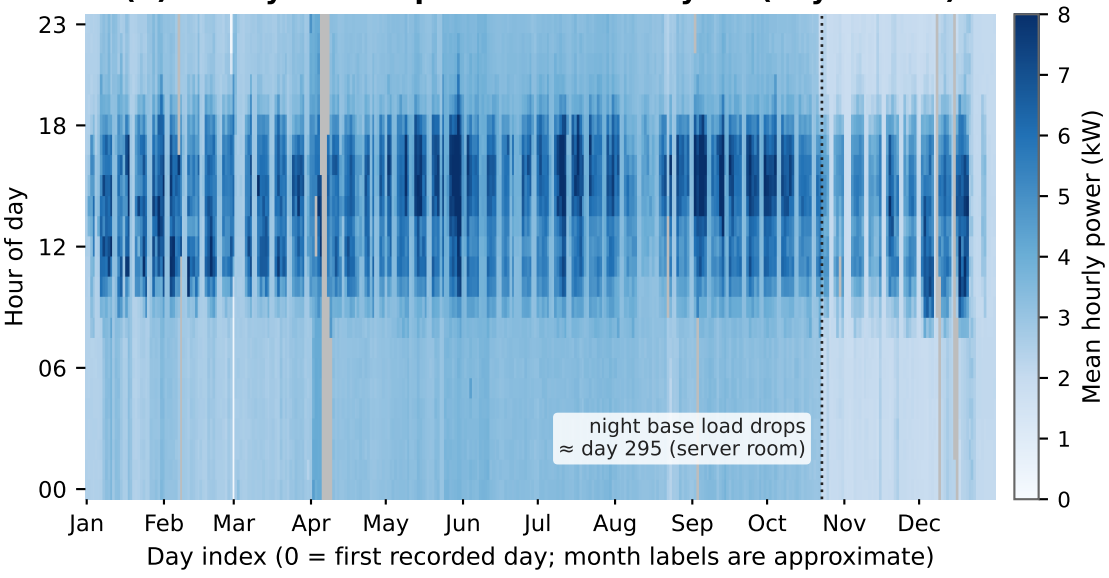
When and where is energy consumed in a smart research building, and how well does PV generation match that rhythm?
Source: Gomes, Pinto & Vale (2022), RETINA/GECAD dataset, 365 days × 288 five-minute intervals.

(a) Daily rhythm: consumption vs PV generation



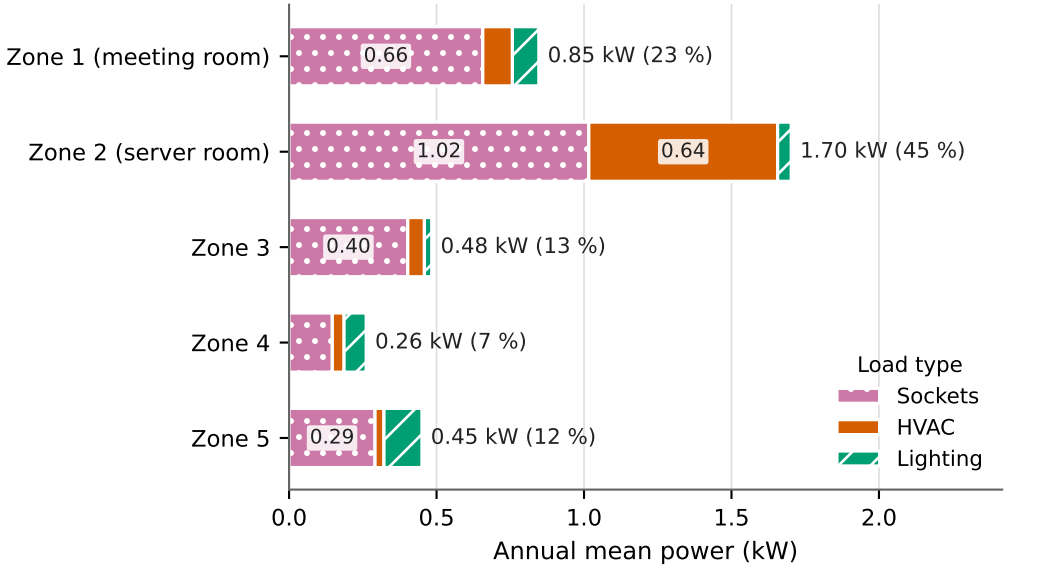
Ribbons = inter-quartile range (25–75 %). Day type inferred from the 7-day periodicity (the dataset contains no dates).

(b) Hourly consumption across the year (day × hour)



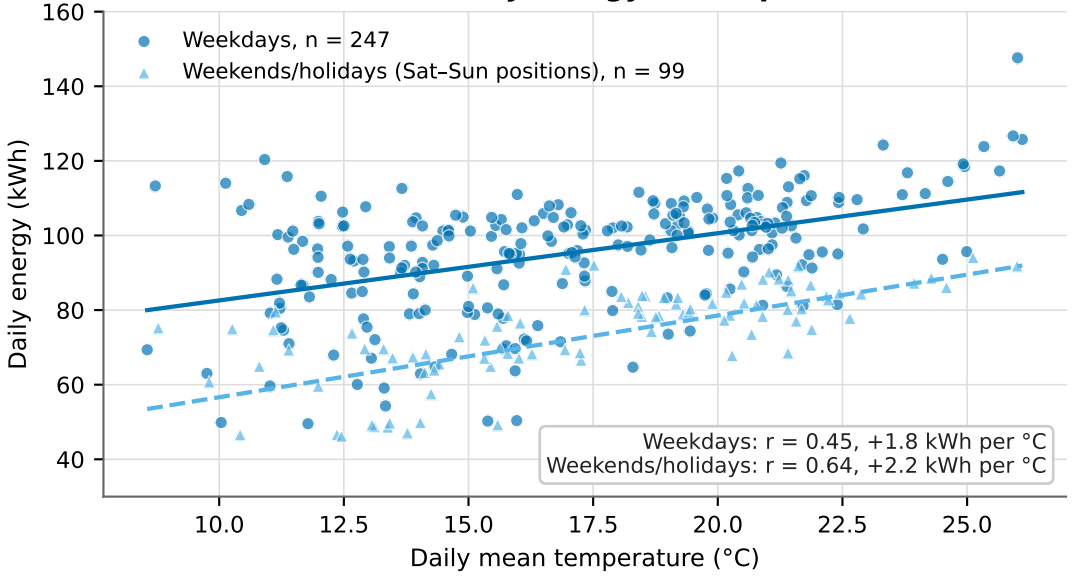
Light vertical columns = weekends & public holidays (7-day rhythm); grey columns = sensor outages (missing values).

(c) Where energy is used: load by zone and type



Full-year averages (including nights & weekends). Sum of zones 3.75 kW ≈ building meter 3.72 kW.

(d) Driver: daily energy vs temperature



Lines = simple linear regression per day type; days with sensor outages excluded.